

3 Predictive control: Preliminaries and notation

Consider a system with **dynamics** described by



$$x[k+1] = Ax[k] + Bu[k] \quad (3.1)$$

The **objective** is to drive the state $x[k]$ to the origin from a given initial condition while respecting state and control constraints in the form

$$x[k] \in \mathcal{X} \quad (3.2)$$

$$u[k] \in \mathcal{U} \quad (3.3)$$



$$\mathcal{U} = \{u \in \mathbb{R}^p : S_u u \leq b_u\} \quad (3.4)$$

and

$$\mathcal{X} = \{x \in \mathbb{R}^n : S_x x \leq b_x\} \quad (3.5)$$



3.1 Cost Function

~~The solution, if it exists, will not be unique. Therefore, it is necessary to adopt a criterion that allows for choosing the best sequence of control actions to perform the task.~~

~~The control problem consists of bringing the states to the origin, so we will employ a cost in the form~~



$$J = \left\| \hat{x}[k+N|k] \right\|_{P_f}^2 + \sum_{i=0}^{N-1} \left[\left\| \hat{x}[k+i|k] \right\|_Q^2 + \left\| \hat{u}[k+i|k] \right\|_R^2 \right] \quad (3.6)$$



and initial condition

$$\hat{x}[k|k] = x[k] \quad (3.7)$$

The expression for J consists of a terminal cost term and a trajectory cost term.

With P_f , Q , R being symmetric and positive-definite, with P_f chosen so that the terminal cost term $\left\| \hat{x}(k+N|k) - \bar{x} \right\|_{P_f}^2$ corresponds to the value function of the infinite horizon Discrete-Time Linear Quadratic Regulator (DLQR).

The cost function can be rewritten using the notation:

$$\hat{\mathbf{u}}[k] = \begin{bmatrix} \hat{u}[k|k] \\ \hat{u}[k+1|k] \\ \vdots \\ \hat{u}[k+N-1|k] \end{bmatrix} \in \mathbb{R}^{pN}, \hat{\mathbf{x}}[k] = \begin{bmatrix} \hat{x}[k+1|k] \\ \hat{x}[k+2|k] \\ \vdots \\ \hat{x}[k+N|k] \end{bmatrix} \in \mathbb{R}^{nN} \quad (3.8)$$

and then the sum equation

$$\begin{aligned} \sum_{j=0}^{N-1} \left\| \hat{u}[j+k|k] \right\|_R^2 &= \hat{u}^T[k|k] R \hat{u}[k|k] + \hat{u}^T[k+1|k] R \hat{u}[k+1|k] + \dots \\ &+ \hat{u}^T[k+N-1|k] R \hat{u}[k+N-1|k] \end{aligned} \quad (3.9)$$

becomes

$$\underbrace{\begin{bmatrix} \hat{u}^T[k|k] & \hat{u}^T[k+1|k] & \dots & \hat{u}^T[k+N-1|k] \end{bmatrix}}_{\hat{\mathbf{u}}^T} \begin{bmatrix} R & 0 & \dots & 0 \\ 0 & R & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & R \end{bmatrix} \underbrace{\begin{bmatrix} \hat{u}[k|k] \\ \hat{u}[k+1|k] \\ \vdots \\ \hat{u}[k+N-1|k] \end{bmatrix}}_{\hat{\mathbf{u}}} \quad (3.10)$$

Using the notation

$$\mathbf{R} = \begin{bmatrix} R & 0 & \cdots & 0 \\ 0 & R & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & R \end{bmatrix} = \text{diag}_N(R) \quad (3.11)$$

we get to

$$\sum_{j=0}^{N-1} \|\hat{u}[j]\|_R^2 = \|\hat{\mathbf{u}}\|_{\mathbf{R}}^2 \quad (3.12)$$

Similarly, we can write

$$\begin{aligned} & \left\| \hat{x}[k+N|k] \right\|_{P_f}^2 + \sum_{j=0}^{N-1} \left\| \hat{x}[k+j|k] \right\|_Q^2 = x^T[k|k] Q x[k|k] \\ & + \hat{x}^T[k+1|k] Q \hat{x}[k+1|k] + \cdots + \hat{x}^T[k+N-1|k] Q \hat{x}[k+N-1|k] + \hat{x}^T[k+N|k] P_f \hat{x}[k+N|k] \\ & = \hat{x}^T[k|k] Q \hat{x}[k|k] + \\ & \underbrace{\begin{bmatrix} \hat{x}^T[k+1|k] & \hat{x}^T[k+2|k] & \cdots & \hat{x}^T[k+N|k] \end{bmatrix}}_{\hat{\mathbf{x}}[k]^T} \underbrace{\begin{bmatrix} Q & 0 & \cdots & 0 \\ 0 & Q & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & P_f \end{bmatrix}}_{\hat{\mathbf{Q}}[k]} \underbrace{\begin{bmatrix} \hat{x}[k+1|k] \\ \hat{x}[k+2|k] \\ \vdots \\ \hat{x}[k+N|k] \end{bmatrix}}_{\hat{\mathbf{x}}[k]} \end{aligned} \quad (3.13)$$

with the notation

$$\mathbf{Q} = \begin{bmatrix} Q & 0 & \cdots & 0 \\ 0 & Q & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & P_f \end{bmatrix} = \begin{bmatrix} \text{diag}_{N-1}(Q) & 0 \\ 0 & P_f \end{bmatrix} \in \mathbb{R}^{nN \times nN} \quad (3.14)$$

as

$$\left\| \hat{x}[k+N|k] \right\|_P^2 + \sum_{j=0}^{N-1} \left\| \hat{x}[k+j|k] \right\|_Q^2 = \left\| x[k] \right\|_P^2 + \left\| \hat{\mathbf{x}} \right\|_{\mathbf{Q}}^2 \quad (3.15)$$

So, the cost function can be written in the form

$$J(\hat{\mathbf{x}}, \hat{\mathbf{u}}) = \left\| x[k] \right\|_P^2 + \left\| \hat{\mathbf{x}} \right\|_{\mathbf{Q}}^2 + \left\| \hat{\mathbf{u}} \right\|_{\mathbf{R}}^2 \quad (3.16)$$

3.2 Constraints

The constraints are introduced aiming to drive the states of the system to the origin while solving

$$\min_{\hat{\mathbf{x}} \in \mathbb{R}^{nN}, \hat{\mathbf{u}} \in \mathbb{R}^{pN}} \mathbf{J}(\hat{\mathbf{x}}, \hat{\mathbf{u}}) \quad (3.17)$$

respecting constraints applied on

$$\hat{\mathbf{u}}[k] = \begin{bmatrix} \hat{u}[k|k] \\ \hat{u}[k+1|k] \\ \vdots \\ \hat{u}[k+N-1|k] \end{bmatrix} \in \mathcal{U} \times \mathcal{U} \times \cdots \times \mathcal{U} = \mathcal{U}^N \quad (3.18)$$

$$\hat{\mathbf{x}}[k] = \begin{bmatrix} \hat{x}[k+1|k] \\ \hat{x}[k+2|k] \\ \vdots \\ \hat{x}[k+N|k] \end{bmatrix} \in \mathcal{X} \times \mathcal{X} \times \cdots \times \mathcal{X}_f = \mathcal{X}^{N-1} \times \mathcal{X}_f \quad (3.19)$$

$\mathcal{X}_f$ is the maximum invariant and admissible set (MAS) concerning the terminal state and control constraints under the action of a terminal control force.

$$\{\hat{x} \in \mathbb{R}^n : S_f \hat{x} \leq b_f\} \quad (3.20)$$

The constraint on $\hat{\mathbf{u}}$ can be written as

$$\begin{bmatrix} S_u & 0 & \cdots & 0 \\ 0 & S_u & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & S_u \end{bmatrix} \begin{bmatrix} \hat{u}[k|k] \\ \hat{u}[k+1|k] \\ \vdots \\ \hat{u}[k+N-1|k] \end{bmatrix} \leq \begin{bmatrix} b_u \\ b_u \\ \vdots \\ b_u \end{bmatrix} \quad (3.21)$$

or, in a shorter form

$$\mathbf{S}_u \hat{\mathbf{u}} \leq \mathbf{b}_u \quad (3.22)$$

with

$$\mathbf{S}_u = \text{diag}_N(S_u), \mathbf{b}_u = \text{col}_N(b_u) \quad (3.23)$$

Similarly, the constraints on $\hat{\mathbf{x}}$ can be written as

$$\begin{bmatrix} S_x & 0 & \cdots & 0 \\ 0 & S_x & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & S_x \end{bmatrix} \begin{bmatrix} \hat{x}[k+1|k] \\ \hat{x}[k+2|k] \\ \vdots \\ \hat{x}[k+N|k] \end{bmatrix} \leq \begin{bmatrix} b_x \\ b_x \\ \vdots \\ b_x \end{bmatrix} \quad (3.24)$$

or, in short

$$\mathbf{S}_x \hat{\mathbf{x}} \leq \mathbf{b}_x \quad (3.25)$$

with

$$\mathbf{S}_x = \begin{bmatrix} \text{diag}_{N-1}(S_x) & 0 \\ 0 & S_f \end{bmatrix}, \mathbf{b}_x = \begin{bmatrix} \text{col}_{N-1}(b_x) \\ b_f \end{bmatrix} \quad (3.26)$$

The vectors $\hat{\mathbf{x}}[k|k]$ and $\hat{\mathbf{u}}[k|k]$ are linked by

$$\hat{\mathbf{x}}[k+1|k] = A\hat{\mathbf{x}}[k|k] + B\hat{\mathbf{u}}[k|k] \quad (3.27)$$

so the constraint



$$\mathbf{S}_x \mathbf{x} \leq \mathbf{b}_x \quad (3.28)$$

can be rewritten as

$$\mathbf{S}_x \mathbf{B} \mathbf{u} \leq \mathbf{b}_x - \mathbf{S}_x \mathbf{A} \mathbf{x}(0) \quad (3.29)$$

Combined with $\mathbf{S}_u \mathbf{u} \leq \mathbf{b}_u$, it can be expressed in the form

$$\mathbf{S} \mathbf{u} \leq \mathbf{b} \quad (3.30)$$

with

$$\mathbf{S} = \begin{bmatrix} \mathbf{S}_x \mathbf{B} \\ \mathbf{S}_u \end{bmatrix}, \mathbf{b} = \begin{bmatrix} \mathbf{b}_x - \mathbf{S}_x \mathbf{A} \hat{\mathbf{x}}[k|k] \\ \mathbf{b}_u \end{bmatrix} \quad (3.31)$$

3.3 Prediction Equation

Let's express the relationship between the vectors

$$\hat{\mathbf{u}}[k] = \begin{bmatrix} \hat{u}[k|k] \\ \hat{u}[k+1|k] \\ \vdots \\ \hat{u}[k+N-1|k] \end{bmatrix}, \hat{\mathbf{x}}[k] = \begin{bmatrix} \hat{x}[k+1|k] \\ \hat{x}[k+2|k] \\ \vdots \\ \hat{x}[k+N|k] \end{bmatrix} \quad (3.32)$$

based on the **state** equation

$$\hat{x}[k+i+1|k] = A\hat{x}[k+i|k] + B\hat{u}[k+i|k], i = 0, 1, \dots, N \quad (3.33)$$

and initial condition $\hat{x}[k|k] = x[k]$.



The solution for the state equation is:

$$\hat{x}[k+i|k] = A^i \hat{x}[k|k] + \sum_{j=0}^{i-1} A^{i-1-j} B \hat{u}[k+j|k] \quad (3.34)$$

and it can be written for $k = 1, 2, \dots, N$ as

$$\hat{x}[k+1|k] = A\hat{x}[k|k] + B\hat{u}[k|k] \quad (3.35)$$

$$\hat{x}[k+2|k] = A^2\hat{x}[k|k] + AB\hat{u}[k|k] + B\hat{u}[k+1|k] \quad (3.36)$$

$$\vdots \quad (3.37)$$

$$\hat{x}[k+N|k] = A^N \hat{x}[k|k] + A^{N-1}B\hat{u}[k|k] + \dots + AB\hat{u}[k+N-2|k] + B\hat{u}[k+N-1|k] \quad (3.38)$$

or

$$\begin{bmatrix} \hat{x}[k+1|k] \\ \hat{x}[k+2|k] \\ \vdots \\ \hat{x}[k+N|k] \end{bmatrix} = \begin{bmatrix} A \\ A^2 \\ \vdots \\ A^N \end{bmatrix} x(0) + \begin{bmatrix} B & 0 & \dots & 0 \\ AB & B & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ A^{N-1}B & A^{N-2}B & \dots & B \end{bmatrix} \begin{bmatrix} \hat{u}[k|k] \\ \hat{u}[k+1|k] \\ \vdots \\ \hat{u}[k+N-1|k] \end{bmatrix} \quad (3.39)$$

This can be written in a shorter form as

$$\hat{\mathbf{x}}[k] = \mathbf{A}x[k] + \mathbf{B}\hat{\mathbf{u}}[k] \quad (3.40)$$

- ajustar notacao $\hat{x}[k]$
- procurar $\hat{x}[k|k] \rightarrow \hat{x}[k]$ (negrito)
- procurar $\hat{u}$ negrito $\rightarrow \hat{u}$ negrito $[k]$
- conferir duplicidade de notacao

with

$$\mathbf{A} = \begin{bmatrix} A \\ A^2 \\ \vdots \\ A^N \end{bmatrix}, \mathbf{B} = \begin{bmatrix} B & 0 & \cdots & 0 \\ AB & B & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ A^{N-1}B & A^{N-2}B & \cdots & B \end{bmatrix} \quad (3.41)$$

3.4 Optimization Problem

We can rewrite the cost function as

$$J = \left\| x[k|k] \right\|_P^2 + \overbrace{\left\| \hat{\mathbf{x}} \right\|_Q^2}^{\square} + \left\| \hat{\mathbf{u}} \right\|_R^2 \quad (3.42)$$

$$\square = \left\| \hat{\mathbf{x}} \right\|_Q^2 \quad (3.43)$$

$$= \left[\mathbf{A}x[k|k] + \mathbf{B}\hat{\mathbf{u}} \right]^T \mathbf{Q} \left[\mathbf{A}x[k|k] + \mathbf{B}\hat{\mathbf{u}} \right] \quad (3.44)$$

$$= \left[\mathbf{A}x[k|k] + \mathbf{B}\hat{\mathbf{u}} \right]^T \mathbf{Q} \left[\mathbf{A}x[k|k] + \mathbf{B}\hat{\mathbf{u}} \right] \quad (3.45)$$

$$= \left\| \mathbf{A}x[k|k] \right\|_Q^2 + \left[\mathbf{A}x[k|k] \right]^T \mathbf{Q} \mathbf{B} \hat{\mathbf{u}} + \hat{\mathbf{u}}^T \mathbf{B}^T \mathbf{Q} \mathbf{A} x[k|k] + \hat{\mathbf{u}}^T \mathbf{B}^T \mathbf{Q} \mathbf{B} \hat{\mathbf{u}} \quad (3.46)$$

replacing $\square$ back in the cost equation and reordering the elements

$$J(\hat{\mathbf{x}}, \hat{\mathbf{u}}) = \left\| x[k|k] \right\|_P^2 + \left\| \mathbf{A}x[k|k] \right\|_Q^2 \quad (3.47)$$

$$+ \left[\mathbf{A}x[k|k] \right]^T \mathbf{Q} \mathbf{B} \hat{\mathbf{u}} + \hat{\mathbf{u}}^T \mathbf{B}^T \mathbf{Q} \mathbf{A} x[k|k] \quad (3.48)$$

$$+ \hat{\mathbf{u}}^T \mathbf{R} \hat{\mathbf{u}} + \hat{\mathbf{u}}^T \mathbf{B}^T \mathbf{Q} \mathbf{B} \hat{\mathbf{u}} \quad (3.49)$$

The constant part will be denoted by c_0

Recalling that the matrix $\mathbf{Q}$ is symmetric, we have:

$$\left[\mathbf{A}x[k|k] \right]^T \mathbf{Q} \mathbf{B} \hat{\mathbf{u}} = \left\{ \hat{\mathbf{u}}^T \mathbf{B}^T \mathbf{Q} \mathbf{A} x[k|k] \right\}^T \quad (3.50)$$

As these terms are scalars, it can be concluded that they are equal to each other, therefore:

$$\begin{aligned} & \left[\mathbf{A}x[k|k] \right]^T \mathbf{Q} \mathbf{B} \hat{\mathbf{u}} + \hat{\mathbf{u}}^T \mathbf{B}^T \mathbf{Q} \mathbf{A} x(0) \\ &= 2 \left[\mathbf{A}x[k|k] \right]^T \mathbf{Q} \mathbf{B} \hat{\mathbf{u}} \end{aligned} \quad (3.51)$$

Finally, it can be written

$$J(\hat{\mathbf{u}}) = \hat{\mathbf{u}}^T (\mathbf{R} + \mathbf{B}^T \mathbf{Q} \mathbf{B}) \hat{\mathbf{u}} + 2 \left[\mathbf{A}x[k|k] \right]^T \mathbf{Q} \mathbf{B} \hat{\mathbf{u}} + c_0 \quad (3.52)$$

with

$$c_0 = \left\| x(0) \right\|_P^2 + \left\| \mathbf{A}x(0) \right\|_Q^2 \quad (3.53)$$

$$c_0 = \left\| x[k|k] \right\|_P^2 + \left\| \mathbf{A}x[k|k] \right\|_Q^2$$

Note that we will now use the notation $J(\hat{\mathbf{u}})$ instead of $J(\hat{\mathbf{x}}, \hat{\mathbf{u}})$, since $\hat{\mathbf{x}}$ has been eliminated from the cost expression.

To finalize the restructuring of the optimization problem in terms of $\hat{\mathbf{u}}$, the relation between $\hat{\mathbf{x}}$ and $\hat{\mathbf{u}}$ in the **restrictions** must be considered.

The optimization problem can be expressed as

$$\min_{\hat{\mathbf{x}} \in \mathbb{R}^{nN}, \hat{\mathbf{u}} \in \mathbb{R}^{pN}} J(\hat{x}, \hat{u}) \quad (3.54)$$

where

$$J(\hat{\mathbf{x}}, \hat{\mathbf{u}}) = \left\| x[k|k] \right\|_P^2 + \left\| \hat{\mathbf{x}} \right\|_Q^2 + \left\| \hat{\mathbf{u}} \right\|_R^2 \quad (3.55)$$

with

$$\hat{\mathbf{u}}[k] = \begin{bmatrix} \hat{u}[k|k] \\ \hat{u}[k+1|k] \\ \vdots \\ \hat{u}[k+N-1|k] \end{bmatrix} \in \mathbb{R}^{pN}, \hat{\mathbf{x}}[k] = \begin{bmatrix} \hat{x}[k+1|k] \\ \hat{x}[k+2|k] \\ \vdots \\ \hat{x}[k+N|k] \end{bmatrix} \in \mathbb{R}^{nN}, \quad (3.56)$$

subject to

$$\mathbf{S}_x \hat{\mathbf{x}}[k] \leq \mathbf{b}_x \quad (3.57)$$

$$\mathbf{S}_u \hat{\mathbf{u}}[k] \leq \mathbf{b}_u \quad (3.58)$$

Thus, the cost can be rewritten as

$$J(\hat{\mathbf{u}}[k]) = \hat{\mathbf{u}}[k]^T (\mathbf{R} + \mathbf{B}^T \mathbf{Q} \mathbf{B}) \hat{\mathbf{u}}[k] + 2 \left[\mathbf{A} \hat{x}[k|k] \right]^T \mathbf{Q} \mathbf{B} \hat{\mathbf{u}}[k] + c_0$$

with

$$c_0 = \left\| \hat{x}[k|k] \right\|_P^2 + \left\| \mathbf{A} \hat{x}[k|k] \right\|_Q^2$$

and the constraints can be expressed as

$$\mathbf{S} \hat{\mathbf{u}}[k] \leq \mathbf{b} \quad (3.59)$$

Having obtained the solution $\hat{\mathbf{u}}[k|k]$ for the problem $\mathbb{P}(\hat{x}[k|k])$, the control to be applied to the plant is given by

$$\hat{u}[k|k] = \hat{u}^*[k|k]$$

where $\hat{u}^*[k|k]$ corresponds to the first p elements of $\hat{\mathbf{u}}^*[k|k]$.

3.5 Feasibility Region

We aim to determine the set $\mathcal{D}$ of points $x[k] \in \mathcal{X}$ for which there exists $\mathbf{u} \in \mathbb{R}^{pN}$ satisfying the constraints

$$\begin{bmatrix} \mathbf{S}_x \mathbf{B} \\ \mathbf{S}_u \end{bmatrix} \mathbf{u} \leq \begin{bmatrix} \mathbf{b}_x - \mathbf{S}_x \mathbf{A} x[k] \\ \mathbf{b}_u \end{bmatrix} \quad (3.60)$$

These restrictions can be rewritten as

$$\begin{bmatrix} \mathbf{S}_x \mathbf{B} & \mathbf{S}_x \mathbf{A} \\ \mathbf{S}_u & x(0) \end{bmatrix} \begin{bmatrix} \mathbf{u} \\ x(0) \end{bmatrix} \leq \begin{bmatrix} \mathbf{b}_x \\ \mathbf{b}_u \end{bmatrix} \quad (3.61)$$

adding a restriction $S_x x(0) \leq b_x$

$$\begin{bmatrix} \mathbf{S}_x \mathbf{B} & \mathbf{S}_x \mathbf{A} \\ \mathbf{S}_u & x(0) \\ 0 & S_x \end{bmatrix} \begin{bmatrix} \mathbf{u} \\ x(0) \end{bmatrix} \leq \begin{bmatrix} \mathbf{b}_x \\ \mathbf{b}_u \\ b_x \end{bmatrix} \quad (3.62)$$

This can be written in a shorter form

$$S_w w \leq b_w \quad (3.63)$$

with

$$w = \begin{bmatrix} \mathbf{u} \\ x(0) \end{bmatrix}, S_w = \begin{bmatrix} \mathbf{S}_x \mathbf{B} & \mathbf{S}_x \mathbf{A} \\ \mathbf{S}_u & 0 \\ 0 & S_x \end{bmatrix}, b_w = \begin{bmatrix} \mathbf{b}_x \\ \mathbf{b}_u \\ b_x \end{bmatrix} \quad (3.64)$$

The constraint $S_w w \leq b_w$ define a polyhedron $\mathcal{P}_w$ in the space $\mathbb{R}^{(pN+n)}$ corresponding to the variables $\mathbf{u}, x(0)$.

The set $\mathcal{D}$ is obtained by projection $\mathcal{P}_w$ onto the subspace of its last n components.

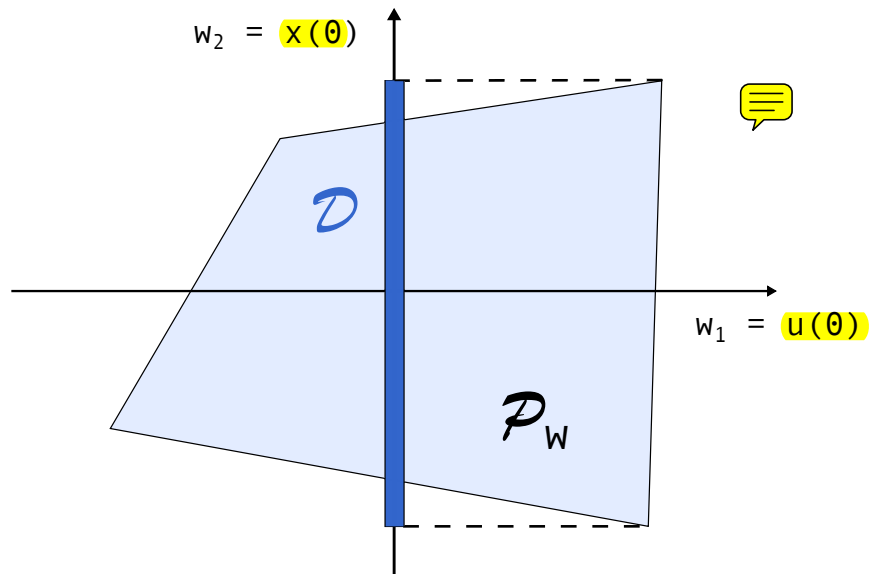


FIGURE 3.1 – Ilustracao para $n = 1$ estado, $p = 1$ entrada e $N = 1$ passo

The **projection** can be performed using the *projection* function from the *MPT Toolbox*.

4 Predictive Control for Stabilization of Periodic Trajectories

4.1 Statement of the problem

Considering a plant of an affine switched system described by:

$$\dot{x}(t) = A_{\sigma(t)}x(t) + b_{\sigma(t)} \quad (4.1)$$

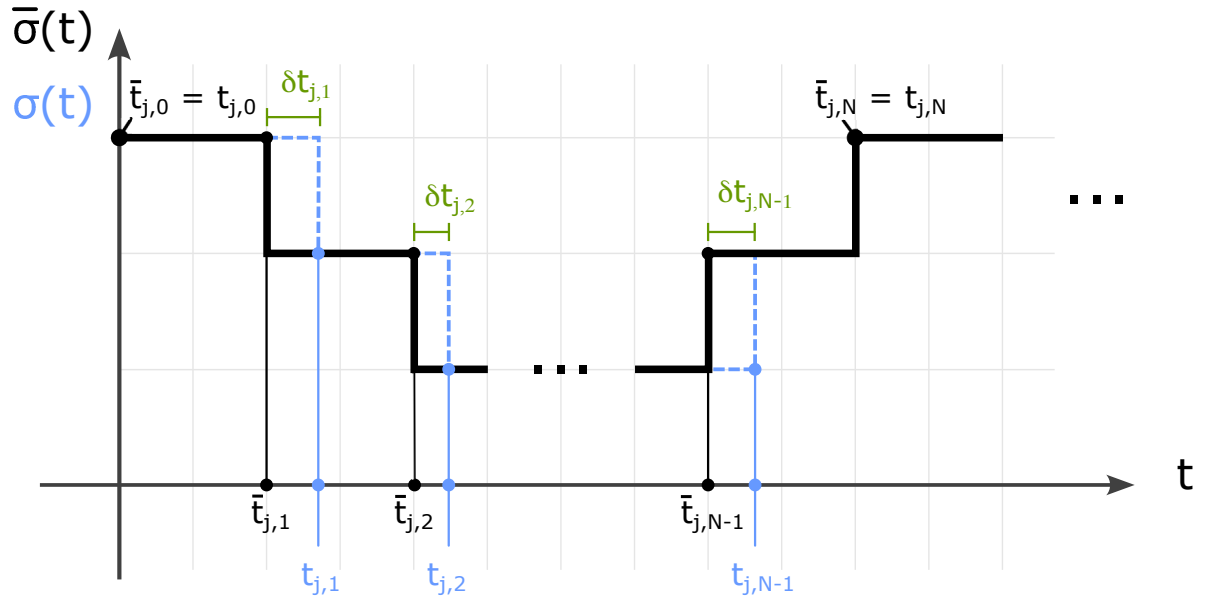


FIGURE 4.1 – Switching signals

 We assume the existence of a periodic switching signal $\bar{\sigma}(t)$ associated with a periodic trajectory $\bar{x}(t)$, as obtained in Chapter 2. Our objective is to introduce perturbations in the switching instants such that the trajectory $x(t)$ converges to the reference trajectory $\bar{x}(t)$.

Figure 4.1 displays the j_{th} cycle of the switching signals $\bar{\sigma}(t)$ and $\sigma(t)$, which comprise N switching actions. Within this cycle, the i_{th} switching times of $\bar{\sigma}(t)$ and $\sigma(t)$ are denoted by $\bar{t}_{j,i}$ and $t_{j,i}$, respectively. The relation between $\bar{t}_{j,i}$ and $t_{j,i}$ is

given by

$$t_{j,i} = \bar{t}_{j,i} + \delta t_{j,i} \quad (4.2)$$



To do this using predictive control, we need to find a linear equation that relates the switching times and the resulting system at the end of one cycle.

4.2 Linearization procedure



The solution for the state equation (4.1) at time t_1 , starting from the initial condition $x(t_0)$, is of the form

$$x(t_1) = e^{A_1(t_1-t_0)}x(t_0) + \int_{t_0}^{t_1} e^{A_1(t_1-\tau)}B_1d\tau \quad (4.3)$$

$$\bar{x}(\bar{t}_1) = e^{A_1(\bar{t}_1-\bar{t}_0)}\bar{x}(\bar{t}_0) + \int_{\bar{t}_0}^{\bar{t}_1} e^{A_1(\bar{t}_1-\tau)}B_1d\tau \quad (4.4)$$

subtracting (4.4) from (4.3) we get

$$\begin{aligned} x(t_1) - \bar{x}(\bar{t}_1) &= \overbrace{e^{A_1(t_1-t_0)}x(t_0) - e^{A_1(\bar{t}_1-\bar{t}_0)}\bar{x}(\bar{t}_0)}^{\square} \\ &\quad + \underbrace{\int_{t_0}^{t_1} e^{A_1(t_1-\tau)}B_1d\tau - \int_{\bar{t}_0}^{\bar{t}_1} e^{A_1(\bar{t}_1-\tau)}B_1d\tau}_{\bigcirc} \end{aligned} \quad (4.5)$$

expanding (4.5) first in $\square$:

$$\begin{aligned} \square &= e^{A_1(t_1-t_0)}x(t_0) - e^{A_1(\bar{t}_1-\bar{t}_0)}\bar{x}(\bar{t}_0) \\ &= e^{A_1(\bar{t}_1+\delta t_1-t_0)}x(t_0) - e^{A_1(\bar{t}_1-\bar{t}_0)}\bar{x}(\bar{t}_0) \\ &= e^{A_1(\bar{t}_1-t_0)}e^{\delta t_1}x(t_0) - e^{A_1(\bar{t}_1-\bar{t}_0)}\bar{x}(\bar{t}_0) \\ &= \bar{F}_1 e^{\delta t_1}x(t_0) - \bar{F}_1 \bar{x}(\bar{t}_0) \\ &= \bar{F}_1 (I + A_1 \delta t_1)x(t_0) - \bar{F}_1 \bar{x}(\bar{t}_0) \\ &= \bar{F}_1 [x(t_0) - \bar{x}(\bar{t}_0)] + \bar{F}_1 A_1 \delta t_1 \end{aligned} \quad (4.6)$$

and then in $\bigcirc$:

$$\begin{aligned}
 \bigcirc &= \int_{t_0}^{t_1} e^{A_1(t_1-\tau)} B_1 d\tau - \int_{\bar{t}_0}^{\bar{t}_1} e^{A_1(\bar{t}_1-\tau)} B_1 d\tau \\
 &= \int_{t_0}^{\bar{t}_1+\delta t_1} e^{A_1(\bar{t}_1+\delta t_1-\tau)} B_1 d\tau - \int_{t_0}^{\bar{t}_1} e^{A_1(\bar{t}_1-\tau)} B_1 d\tau \\
 &= e^{A_1\delta t_1} \left[\int_{t_0}^{\bar{t}_1} e^{A_1(\bar{t}_1-\tau)} B_1 d\tau + \int_{\bar{t}_1}^{\bar{t}_1+\delta t_1} e^{A_1(t_1-\tau)} B_1 d\tau \right] - \int_{t_0}^{\bar{t}_1} e^{A_1(\bar{t}_1-\tau)} B_1 d\tau \\
 &= e^{A_1\delta t_1} [\bar{G}_1 + B_1\delta t_1] - \bar{G}_1 \\
 &= (I + A_1\delta t_1)[\bar{G}_1 + B_1\delta t_1] - \bar{G}_1 \\
 &= \cancel{\bar{G}_1} + B_0\delta t_1 + A_1\bar{G}_1\delta t_1 + \cancel{A_1B_1\delta t_1\delta t_1} - \cancel{\bar{G}_1} \\
 &= A_1\bar{G}_1\delta t_1 + B_1\delta t_1
 \end{aligned} \tag{4.7}$$

the resulting equations (4.6) and (4.7) are replaced back in (4.5) to complete the difference equation

$$x(t_1) - \bar{x}(\bar{t}_1) = \bar{F}_1[x(t_0) - \bar{x}(\bar{t}_0)] + [\bar{F}_1A_1x(t_0) + A_1\bar{G}_1 + B_1]\delta t_1 \tag{4.8}$$

Repeating the steps for $x(t_2)$ we have

$$x(t_2) = e^{A_2(t_2-t_1)}x(t_1) + \int_{t_1}^{t_2} e^{A_2(t_2-\tau)} B_2 d\tau \tag{4.9}$$

$$\bar{x}(\bar{t}_2) = e^{A_2(\bar{t}_2-\bar{t}_1)}\bar{x}(\bar{t}_1) + \int_{\bar{t}_1}^{\bar{t}_2} e^{A_2(\bar{t}_2-\tau)} B_2 d\tau \tag{4.10}$$

subtracting (4.10) from (4.9)

$$\begin{aligned}
 x(t_2) - \bar{x}(\bar{t}_2) &= \overbrace{e^{A_2(t_2-t_1)}x(t_1) - e^{A_2(\bar{t}_2-\bar{t}_1)}\bar{x}(\bar{t}_1)}^{\square} \\
 &\quad + \underbrace{\int_{t_1}^{t_2} e^{A_2(t_2-\tau)} B_2 d\tau - \int_{\bar{t}_1}^{\bar{t}_2} e^{A_2(\bar{t}_2-\tau)} B_2 d\tau}_{\bigcirc}
 \end{aligned} \tag{4.11}$$

expanding $\square$ in (4.11):

$$\begin{aligned}
 \square &= e^{A_2(\bar{t}_2+\delta t_2-\bar{t}_1-\delta t_1)}x(t_1) - e^{A_2(\bar{t}_2-\bar{t}_1)}\bar{x}(\bar{t}_1) \\
 &= e^{A_2(\bar{t}_2-\bar{t}_1)}e^{A_2(\delta t_2-\delta t_1)}x(t_1) - e^{A_2(\bar{t}_2-\bar{t}_1)}\bar{x}(\bar{t}_1) \\
 &= \bar{F}_2[I + A_2(\delta t_2 - \delta t_1)]x(t_1) - \bar{F}_2\bar{x}(\bar{t}_1) \\
 &= \bar{F}_2[x(t_1) - \bar{x}(\bar{t}_1)] + \bar{F}_2A_2x(t_1)\delta t_2 - \bar{F}_2A_2x(t_1)\delta t_1
 \end{aligned} \tag{4.12}$$

and expanding $\bigcirc$:

$$\begin{aligned}
 \bigcirc &= \int_{t_1}^{t_2} e^{A_2(t_2-\tau)} B_2 d\tau - \int_{\bar{t}_1}^{\bar{t}_2} e^{A_2(\bar{t}_2-\tau)} B_2 d\tau \\
 &= \int_{\bar{t}_1+\delta t_1}^{\bar{t}_2+\delta t_2} e^{A_2(\bar{t}_2+\delta t_2-\tau)} B_2 d\tau - \int_{\bar{t}_1}^{\bar{t}_2} e^{A_2(\bar{t}_2-\tau)} B_2 d\tau \\
 &= e^{A_2\delta t_2} \left[\int_{\bar{t}_1}^{\bar{t}_2} e^{A_2(\bar{t}_2-\tau)} B_2 d\tau - \int_{\bar{t}_1}^{\bar{t}_1+\delta t_1} e^{A_2(\bar{t}_2-\tau)} B_2 d\tau + \int_{\bar{t}_2}^{\bar{t}_2+\delta t_2} e^{A_2(\bar{t}_2-\tau)} B_2 d\tau \right] \\
 &\quad - \int_{\bar{t}_1}^{\bar{t}_2} e^{A_2(\bar{t}_2-\tau)} B_2 d\tau \\
 &= (I + A_2\delta t_2) \left[\bar{G}_2 - \int_{\bar{t}_1}^{\bar{t}_1+\delta t_1} e^{A_2(\bar{t}_2-\tau)} B_2 d\tau + \int_{\bar{t}_2}^{\bar{t}_2+\delta t_2} e^{A_2(\bar{t}_2-\tau)} B_2 d\tau \right] - \bar{G}_2 \\
 &= (I + A_2\delta t_2) \left[\bar{G}_2 - e^{A_2(\bar{t}_2-\bar{t}_1)} B_2 \delta t_1 + B_2 \delta t_2 \right] - \bar{G}_2 \\
 &= (I + A_2\delta t_2) \left[\bar{G}_2 - \bar{F}_2 B_2 \delta t_1 + B_2 \delta t_2 \right] - \bar{G}_2 \\
 &= \cancel{\bar{G}_2} - \bar{F}_2 B_2 \delta t_1 + B_2 \delta t_2 + A_2 \bar{G}_2 \delta t_2 - \cancel{\bar{G}_2} \\
 &= A_2 \bar{G}_2 \delta t_2 + B_2 \delta t_2 - \bar{F}_2 B_2 \delta t_1
 \end{aligned} \tag{4.13}$$

and replacing (4.12) and (4.13) back into equation (4.11) we get to the difference equation for $x(t_2) - \bar{x}(\bar{t}_2)$

$$\begin{aligned}
 x(t_2) - \bar{x}(\bar{t}_2) &= \bar{F}_2 [x(t_1) - \bar{x}_1(\bar{t}_1)] \\
 &\quad - \bar{F}_2 [A_2 x(t_1) + B_2] \delta t_1 \\
 &\quad + [\bar{F}_2 A_2 \underbrace{x(t_1)}_{\triangle} + A_2 \bar{G}_2 + B_2] \delta t_2
 \end{aligned} \tag{4.14}$$

from (4.14), $\triangle$ can be expanded by replacing the resulting equation of (4.8)

$$\triangle = x(t_1) = \bar{x}(\bar{t}_1) + \bar{F}_1 [x(t_0) - \bar{x}(t_0)] + [\bar{F}_1 A_1 x(t_0) + A_1 \bar{G}_1 + B_1] \delta t_1 \tag{4.15}$$

note that when $\triangle$ is replaced in eq. (4.14) it will produce a product with δt_1 and δt_2 like

$$x(t_1) \delta t_1 = \bar{x}(\bar{t}_1) \delta t_1 + \bar{F}_1 \overbrace{[x(t_0) - \bar{x}(t_0)]}^{\square} \delta t_1 + [\bar{F}_1 A_1 x(t_0) + A_1 \bar{G}_1 + B_1] \overbrace{\delta t_1 \delta t_1}^{\bigcirc}$$

both $\square$ and $\bigcirc$ are products of differences that produces a smaller number and they will be neglected in the first order approximations

$$x(t_1) \delta t_1 \doteq \bar{x}(\bar{t}_1) \delta t_1 \tag{4.16}$$

replacing (4.8) and (4.15) in (4.14) it follows that

$$\begin{aligned}
 x(t_2) - \bar{x}(\bar{t}_2) &= \bar{F}_2 \{ \bar{F}_1 [x(t_0) - \bar{x}(t_0) + [\bar{F}_1 A_1 x(t_0) + A_1 \bar{G}_1 + B_1] \delta t_1] \} \\
 &\quad - \bar{F}_2 [A_2 \bar{x}(t_1) + B_2] \delta t_1 \\
 &\quad + [\bar{F}_2 A_2 \bar{x}(t_1) + A_2 \bar{G}_2 + B_2] \delta t_2
 \end{aligned}$$

$$\begin{aligned}
 x(t_2) - \bar{x}(\bar{t}_2) &= \bar{F}_2 \bar{F}_1 [x(t_0) - \bar{x}(t_0)] \\
 &\quad + \bar{F}_2 [\bar{F}_1 A_1 x(t_0) + A_1 \bar{G}_1 - A_2 \bar{x}(\bar{t}_1) + B_1 - B_2] \delta t_1 \\
 &\quad + [\bar{F}_2 A_2 \bar{x}(\bar{t}_1) + A_2 \bar{G}_2 + B_2] \delta t_2
 \end{aligned} \tag{4.17}$$

Repeating the procedure for $x(t_3)$ we have

$$x(t_3) = e^{A_3(t_3-t_2)} x(t_2) + \int_{t_2}^{t_3} e^{A_3(t_3-\tau)} B_3 d\tau \tag{4.18}$$

$$\bar{x}(\bar{t}_3) = e^{A_3(\bar{t}_3-\bar{t}_2)} \bar{x}(\bar{t}_2) + \int_{\bar{t}_2}^{\bar{t}_3} e^{A_3(\bar{t}_3-\tau)} B_3 d\tau \tag{4.19}$$

subtracting (4.18) from (4.19)

$$\begin{aligned}
 x(t_3) - \bar{x}(\bar{t}_3) &= \overbrace{e^{A_3(t_3-t_2)} x(t_2) - e^{A_3(\bar{t}_3-\bar{t}_2)} \bar{x}(\bar{t}_2)}^{\square} \\
 &\quad + \underbrace{\int_{t_2}^{t_3} e^{A_3(t_3-\tau)} B_3 d\tau - \int_{\bar{t}_2}^{\bar{t}_3} e^{A_3(\bar{t}_3-\tau)} B_3 d\tau}_{\bigcirc}
 \end{aligned} \tag{4.20}$$

expanding $\square$ in (4.20):

$$\begin{aligned}
 \square &= e^{A_3(\bar{t}_3+\delta t_3-\bar{t}_2-\delta t_2)} x(t_2) - e^{A_3(\bar{t}_3-\bar{t}_2)} \bar{x}(\bar{t}_2) \\
 &= e^{A_3(\bar{t}_3-\bar{t}_2)} e^{A_3(\delta t_3-\delta t_2)} x(t_2) - e^{A_3(\bar{t}_3-\bar{t}_2)} \bar{x}(\bar{t}_2) \\
 &= \bar{F}_3 [I + A_3(\delta t_3 - \delta t_2)] x(t_2) - \bar{F}_3 \bar{x}(\bar{t}_2) \\
 &= \bar{F}_3 [x(t_2) - \bar{x}(\bar{t}_2)] + \bar{F}_3 A_3 x(t_2) \delta t_3 - \bar{F}_3 A_3 x(t_2) \delta t_2
 \end{aligned} \tag{4.21}$$

and expanding $\bigcirc$:

$$\begin{aligned}
 \bigcirc &= \int_{t_2}^{t_3} e^{A_3(t_3-\tau)} B_3 d\tau - \int_{\bar{t}_2}^{\bar{t}_3} e^{A_3(\bar{t}_3-\tau)} B_3 d\tau \\
 &= \int_{\bar{t}_2+\delta t_2}^{\bar{t}_3+\delta t_3} e^{A_3(\bar{t}_3+\delta t_3-\tau)} B_3 d\tau - \int_{\bar{t}_2}^{\bar{t}_3} e^{A_3(\bar{t}_3-\tau)} B_3 d\tau \\
 &= e^{A_3\delta t_3} \left[\int_{\bar{t}_2}^{\bar{t}_3} e^{A_3(\bar{t}_3-\tau)} B_3 d\tau - \int_{\bar{t}_2}^{\bar{t}_2+\delta t_2} e^{A_3(\bar{t}_3-\tau)} B_3 d\tau + \int_{\bar{t}_3}^{\bar{t}_3+\delta t_3} e^{A_3(\bar{t}_3-\tau)} B_3 d\tau \right] \\
 &\quad - \int_{\bar{t}_2}^{\bar{t}_3} e^{A_3(\bar{t}_3-\tau)} B_3 d\tau \\
 &= (I + A_3\delta t_3) \left[\bar{G}_3 - \int_{\bar{t}_2}^{\bar{t}_2+\delta t_2} e^{A_3(\bar{t}_3-\tau)} B_3 d\tau + \int_{\bar{t}_3}^{\bar{t}_3+\delta t_3} e^{A_3(\bar{t}_3-\tau)} B_3 d\tau \right] - \bar{G}_3 \\
 &= (I + A_3\delta t_3) \left[\bar{G}_3 - e^{A_3(\bar{t}_3-\bar{t}_2)} B_3 \delta t_2 + B_3 \delta t_3 \right] - \bar{G}_3 \\
 &= (I + A_3\delta t_3) \left[\bar{G}_3 - \bar{F}_3 B_3 \delta t_2 + B_3 \delta t_3 \right] - \bar{G}_3 \\
 &= \cancel{\bar{G}_3} - \bar{F}_3 B_3 \delta t_2 + B_3 \delta t_3 + A_3 \bar{G}_3 \delta t_3 - \cancel{\bar{G}_3} \\
 &= A_3 \bar{G}_3 \delta t_3 + B_3 \delta t_3 - \bar{F}_3 B_3 \delta t_2
 \end{aligned} \tag{4.22}$$

and then replacing (4.21) in (4.22) back in eq. (4.20) we get to the difference equation for $x(t_3) - \bar{x}(\bar{t}_3)$

$$\begin{aligned}
 x(t_3) - \bar{x}(\bar{t}_3) &= \bar{F}_3 [x(t_2) - \bar{x}(\bar{t}_2)] \\
 &\quad - \bar{F}_3 [A_3 x(t_2) + B_3] \delta t_2 \\
 &\quad + [\bar{F}_3 A_3 x(t_2) + A_3 \bar{G}_3 + B_3] \delta t_3
 \end{aligned} \tag{4.23}$$

from (4.17), $\triangle$ can be expanded the same way as in the past iteration and $x(t_2) - \bar{x}(\bar{t}_2)$ can be replaced in (4.23)

$$\begin{aligned}
 x(t_3) - \bar{x}(\bar{t}_3) &= \bar{F}_3 \{ \bar{F}_2 \bar{F}_1 [x(t_0) - \bar{x}(t_0)] \\
 &\quad + \bar{F}_2 [\bar{F}_1 A_1 x(t_0) + A_1 \bar{G}_1 - A_2 \bar{x}(\bar{t}_1) + B_1 - B_2] \delta t_1 \\
 &\quad + [\bar{F}_2 A_2 \bar{x}(t_1) + A_2 \bar{G}_2 + B_2] \delta t_2 \} \\
 &\quad - \bar{F}_3 [A_3 \bar{x}(t_2) + B_3] \delta t_2 \\
 &\quad + [\bar{F}_3 A_3 \bar{x}(t_2) + A_3 \bar{G}_3 + B_3] \delta t_3
 \end{aligned}$$

after rearranging the terms, one arrives at

$$\begin{aligned}
 x(t_3) - \bar{x}(\bar{t}_3) = & \bar{F}_3 \bar{F}_2 \bar{F}_1 [x(t_0) - \bar{x}(t_0)] \\
 & + \bar{F}_3 \bar{F}_2 [\bar{F}_1 A_1 x(t_0) + A_1 \bar{G}_1 - A_2 \bar{x}(\bar{t}_1) + B_1 - B_2] \delta t_1 \\
 & + \bar{F}_3 [\bar{F}_2 A_2 \bar{x}(t_1) + A_2 \bar{G}_2 - A_3 \bar{x}(\bar{t}_2) + B_2 - B_3] \delta t_2 \\
 & + [\bar{F}_3 A_3 \bar{x}(t_2) + A_3 \bar{G}_3 + B_3] \delta t_3
 \end{aligned} \tag{4.24}$$

Now, there are enough iterations to deduce a pattern on each increment of the equation, so the equation for $x(t_i)$ can be deduced as

$$\begin{aligned}
 x(t_i) - \bar{x}(\bar{t}_i) = & \bar{F}_i \bar{F}_{i-1} \dots \bar{F}_1 [x(t_0) - \bar{x}(t_0)] \\
 & + \bar{F}_i \bar{F}_{i-1} \dots \bar{F}_1 [\bar{F}_1 A_1 x(t_0) + A_1 \bar{G}_1 - A_2 \bar{x}(\bar{t}_1) + B_1 - B_2] \delta t_1 \\
 & + \bar{F}_i \bar{F}_{i-1} \dots \bar{F}_2 [\bar{F}_2 A_2 x(t_1) + A_2 \bar{G}_2 - A_3 \bar{x}(\bar{t}_2) + B_2 - B_3] \delta t_2 \\
 & \vdots \\
 & + \bar{F}_i [\bar{F}_{i-1} A_{i-1} \bar{x}(t_{i-2}) + A_{i-1} \bar{G}_{i-1} - A_i \bar{x}(\bar{t}_{i-1}) + B_{i-1} - B_i] \delta t_{i-1} \\
 & + [\bar{F}_i A_i \bar{x}(t_i) + A_i \bar{G}_i + B_i] \delta t_i
 \end{aligned}$$

(4.25)

and then, for $x(t_N)$ the equation is as follows

$$\begin{aligned}
 x(t_N) - \bar{x}(\bar{t}_N) = & \bar{F}_N \bar{F}_{N-1} \dots \bar{F}_1 [x(t_0) - \bar{x}(t_0)] \\
 & + \bar{F}_N \bar{F}_{N-1} \dots \bar{F}_2 \overbrace{[\bar{F}_1 A_1 x(t_0) + A_1 \bar{G}_1 - A_2 \bar{x}(\bar{t}_1) + B_1 - B_2]}^{\square} \delta t_1 \\
 & + \bar{F}_N \bar{F}_{N-1} \dots \bar{F}_3 [\bar{F}_2 A_2 x(t_1) + A_2 \bar{G}_2 - A_3 \bar{x}(\bar{t}_2) + B_2 - B_3] \delta t_2 \\
 & \vdots \\
 & + \bar{F}_N [\bar{F}_{N-1} A_{N-1} \bar{x}(t_{N-2}) + A_{N-1} \bar{G}_{N-1} - A_N \bar{x}(\bar{t}_{N-1}) \\
 & \quad + B_{N-1} - B_N] \delta t_{N-1} \\
 & + \underbrace{[\bar{F}_N A_{N-1} \bar{x}(t_{N-1}) + A_N \bar{G}_N + B_N]}_{\bigcirc} \delta t_N
 \end{aligned} \tag{4.26}$$

The equation (4.26) can be developed further to remove the dependencies on the in-

termediary states $x(t_0), x(t_1), \dots, x(t_{N-2})$. By expanding $\square$ and $\bigcirc$, the following identity holds

$$\begin{aligned}
 \square &= \bar{F}_1 A_1 x(t_0) + A_1 \bar{G}_1 - A_2 \bar{x}(\bar{t}_1) + B_1 - B_2 \\
 &= A_1 \bar{F}_1 x(t_0) + A_1 \bar{G}_1 - A_2 \bar{x}(\bar{t}_1) + B_1 - B_2 \\
 &= A_1 [\bar{x}(\bar{t}_1) - \bar{G}_1] + A_1 \bar{G}_1 - A_2 \bar{x}(\bar{t}_1) + B_1 - B_2 \\
 &= A_1 \bar{x}(\bar{t}_1) - \cancel{A_1 \bar{G}_1} + \cancel{A_1 \bar{G}_1} - A_2 \bar{x}(\bar{t}_1) + B_1 - B_2 \\
 &= (A_1 - A_2) \bar{x}(\bar{t}_1) + B_1 - B_2
 \end{aligned} \tag{4.27}$$

$$\begin{aligned}
 \bigcirc &= \bar{F}_N A_N \bar{x}(t_{N-1}) + A_N \bar{G}_N + B_N \\
 &= A_N \bar{F}_N \bar{x}(t_{N-1}) + A_N \bar{G}_N + B_N \\
 &= A_N [\bar{x}(\bar{t}_N) - \bar{G}_N] + A_N \bar{G}_N + B_N \\
 &= A_N \bar{x}(\bar{t}_N) - \cancel{A_N \bar{G}_N} + \cancel{A_N \bar{G}_N} + B_N \\
 &= A_N \bar{x}(\bar{t}_N) + B_N
 \end{aligned} \tag{4.28}$$

replacing (4.27) and (4.28) in (4.26)

$$\begin{aligned}
 x(t_N) - \bar{x}(\bar{t}_N) &= \bar{F}_N \bar{F}_{N-1} \dots \bar{F}_1 [x(t_0) - \bar{x}(t_0)] \\
 &\quad + \bar{F}_N \bar{F}_{N-1} \dots \bar{F}_2 [(A_1 - A_2) \bar{x}(\bar{t}_1) + B_1 - B_2] \delta t_1 \\
 &\quad + \bar{F}_N \bar{F}_{N-1} \dots \bar{F}_3 [(A_2 - A_3) \bar{x}(\bar{t}_2) + B_2 - B_3] \delta t_2 \\
 &\quad \vdots \\
 &\quad + \bar{F}_N [(A_{N-1} - A_N) \bar{x}(\bar{t}_{N-1}) + B_{N-1} - B_N] \delta t_{N-1} \\
 &\quad + [A_N \bar{x}(\bar{t}_N) + B_N] \delta t_N
 \end{aligned}$$

(4.29)

and renaming the terms of (4.29) to

$$e(t) = x(t) - \bar{x}(\bar{t}) \quad (4.30)$$

$$\Phi = \bar{F}_N \bar{F}_{N-1} \dots \bar{F}_1 \quad (4.31)$$

$$\begin{aligned} \Gamma = & \begin{bmatrix} \bar{F}_N \bar{F}_{N-1} \dots \bar{F}_2 [(A_1 - A_2)\bar{x}(\bar{t}_1) + B_1 - B_2], \\ \bar{F}_N \bar{F}_{N-1} \dots \bar{F}_3 [(A_2 - A_3)\bar{x}(\bar{t}_2) + B_2 - B_3], \\ \vdots \\ \bar{F}_N [(A_{N-1} - A_N)\bar{x}(\bar{t}_{N-1}) + B_{N-1} - B_N], \\ A_N \bar{x}(\bar{t}_N) + B_N \end{bmatrix} \end{aligned} \quad (4.32)$$

$$\delta t = [\delta t_1, \delta t_2, \dots, \delta t_N]^T \quad (4.33)$$

we get to the affine equation

$$e(t_N) = \Phi e(t_0) + \Gamma \delta t \quad (4.34)$$

$$e(t_{j,N}) = \Phi e(t_{j,0}) + \Gamma \delta t[j] \quad (4.35)$$

where $\bar{x}$ is calculated as

$$\bar{x}(\bar{t}_1) = F_1 \bar{x}(\bar{t}_0) + G_1 \quad (4.36)$$

$$\bar{x}(\bar{t}_2) = F_2 \bar{x}(\bar{t}_1) + G_2 \quad (4.37)$$

$$\vdots$$

$$\bar{x}(\bar{t}_N) = F_N \bar{x}(\bar{t}_{N-1}) + G_N \quad (4.38)$$

and F_i and G_i are calculated as

$$[F_i, G_i] = c2dm(Ai, Bi, [], [], dti, 'zoh') \quad (4.39)$$



4.3 MPC Problem

Note that the equation (4.34) relates to the equation (3.1) as both describe the evolution of a system over time. The equation (3.1) represents the system's state and input at a given time step, and the equation (4.34) describes the error between the system's

predicted and actual values at two different time points.

Referring to equation (3.1), we can make some substitutions to simplify our discussion. We'll replace variable x with e , and u with δt . Additionally, we'll denote the cycle index with k . With these substitutions in place, we arrive at equation (4.34).

4.3.1 Cost Function



$$J = \|e[N_p]\|_{P_f}^2 + \sum_{j=0}^{N_p-1} \|e[j]\|_Q^2 + \|\delta t[j]\|_R^2 \quad (4.40)$$

$$e(t_{j,N}) = \Phi e(t_{j,0}) + \Gamma \delta t[j] \quad (4.41)$$

where $Q = Q^T > 0$, $\rho > 0$

$$\delta t[j] = [t_{j,1} - \bar{t}_{j,1}, t_{j,2} - \bar{t}_{j,2}, \dots, t_{j,N-1} - \bar{t}_{j,N-1}]^T \quad (4.42)$$

4.3.2 Dwell-time Constraints

We want to enforce a constraint on the dwell times where the switching times will be greater than a minimum value



$$(t_i - t_{i-1}) \geq \Delta t_{i,min}, \quad i = 1, 2, \dots, N \quad (4.43)$$

we can write this as

$$\delta t_1 \geq -\Delta \bar{t}_1 + \Delta t_{1,min} \quad (4.44)$$

$$\delta t_i - \delta t_{i-1} \geq -\Delta \bar{t}_i + \Delta t_{i,min}, \quad i = 1, 2, \dots, N \quad (4.45)$$

$$\delta t_{N-1} \geq -\Delta \bar{t}_N + \Delta t_{N,min} \quad (4.46)$$

with

$$\Delta \bar{t}_i = \bar{t}_i - \bar{t}_{i-1} \quad (4.47)$$

Rearranging this to a matrix form we have

$$L \delta t \geq c \quad (4.48)$$

with

$$L = \begin{bmatrix} 1 & 0 & 0 & \cdots & 0 & 0 \\ -1 & 1 & 0 & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \\ 0 & 0 & 0 & \cdots & -1 & 1 \\ 0 & 0 & 0 & \cdots & 0 & -1 \end{bmatrix} \quad (4.49)$$

$$\delta t = \begin{bmatrix} \delta t_1 \\ \delta t_2 \\ \vdots \\ \delta t_{N-1} \\ \delta t_N \end{bmatrix} \quad (4.50)$$

$$c = \begin{bmatrix} -\Delta \bar{t}_1 + \Delta t_{1,min} \\ -\Delta \bar{t}_2 + \Delta t_{2,min} \\ \vdots \\ -\Delta \bar{t}_{N-1} + \Delta t_{N-1,min} \\ -\Delta \bar{t}_N + \Delta t_{N,min} \end{bmatrix} \quad (4.51)$$

To use this with the constraints $S\hat{u}[k] \leq b$, this can be rearranged as

$$-L\delta t \leq -c$$

4.4 Application Examples

4.4.1 Double Integrator

4.4.1.1 Parameters of the Simulation

This example describes the double integrator model used in Marcolino (2018) of the form

$$\dot{x}(t) = A_x x(t) + B_c u(t) \quad (4.52)$$

where

$$A_c = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, B_c = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \quad (4.53)$$

and the associated input signal is

$$u_r = \begin{bmatrix} 1 \\ 0 \\ -1 \\ 0 \end{bmatrix}, T_s = [0 \quad 2 \quad 3 \quad 5 \quad 6] \quad (4.54)$$

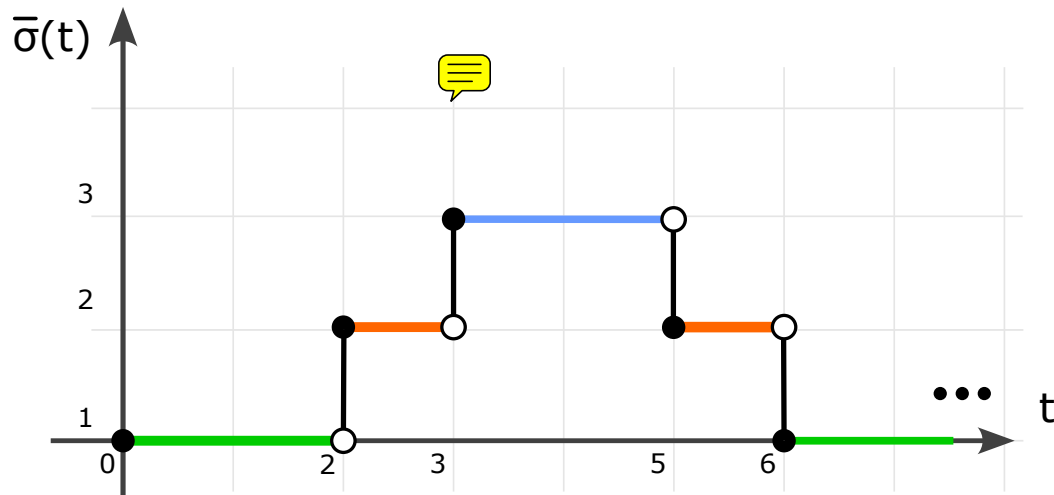


FIGURE 4.2 – Reference signal for Double Integrator

The original problem has the same dynamics during all operations, so we need to construct the associated A_i, b_i for each operation mode. One solution is to create 3 unique pairs

$$A_1 = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \quad A_2 = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \quad A_3 = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$$

$$b_1 = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \quad b_2 = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \quad b_3 = \begin{bmatrix} 0 \\ -1 \end{bmatrix}$$

and the index for the operation mode $\Omega = [1, 2, 3, 2]$, with reference state vector for this trajectory is $\bar{x}_{ref} = [2, -1]$, the maximum value for the period of the trajectory cycle $T_{pmax} = 1$ and initial state of the nominal target trajectory $x_0 = [1.5, -1]$.

4.4.1.2 Prediction Matrices

From (4.32) and (4.33) we construct

$$\Phi = \begin{bmatrix} 1 & 6 \\ 0 & 1 \end{bmatrix} \quad (4.55)$$

$$\Gamma = \begin{bmatrix} 4 & 3 & -1 \\ 1 & 1 & -1 \end{bmatrix} \quad (4.56)$$

with the switching time constraints

$$c = \begin{bmatrix} -\Delta t_{r_1} + \Delta t_{r_1, min} \\ -\Delta t_{r_2} + \Delta t_{r_2, min} \\ \vdots \\ -\Delta t_{r_{N-1}} + \Delta t_{r_{N-1}, min} \\ -\Delta t_{r_N} + \Delta t_{r_N, min} \end{bmatrix} = \begin{bmatrix} -2 + 1.625 \\ -1 + 0.635 \\ -2 + 1.625 \\ -1 + 0.625 \end{bmatrix} = \begin{bmatrix} -0.375 \\ -0.365 \\ -0.375 \\ -0.365 \end{bmatrix} \quad (4.57)$$

4.4.1.3 Results

- FALTA SIMULAR A REGIAO DE FACTIBILIDADE

The simulation uses the system (4.53) with the command signal (4.54) and with a initial condition $x_0 = [1.5, -0.5]$. Figure 4.3 shows the simulation without the MPC control being applied running for 5 cycles and we can see that the trajectory is diverging from the initial position of the target cyclic trajectory $\bar{x}_0 = [-0.5, -1.0]$ position.

Now, for the same condition, we can turn on the MPC controller and run the simulation for 50 cycles. Figure 4.4 shows the resulting trajectory and Figure 4.5 shows the associated command signal. We can see that the controller drives the trajectory of the system to the

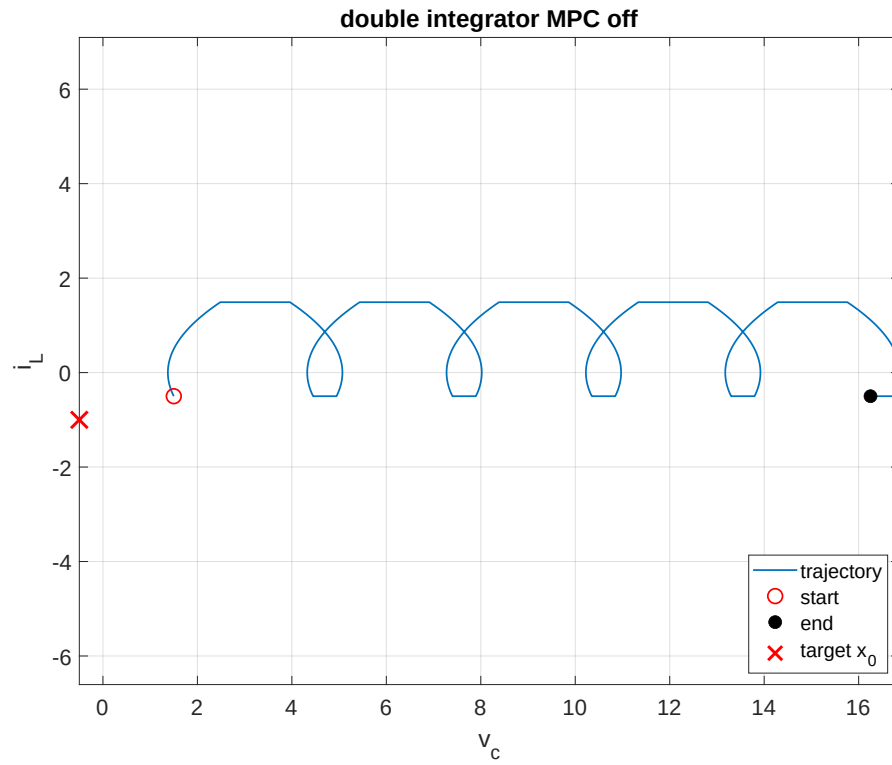


FIGURE 4.3 – State Trajectory MPC off, 4 cycles

target cyclic trajectory, we also that the command signal get in sync with the reference command.

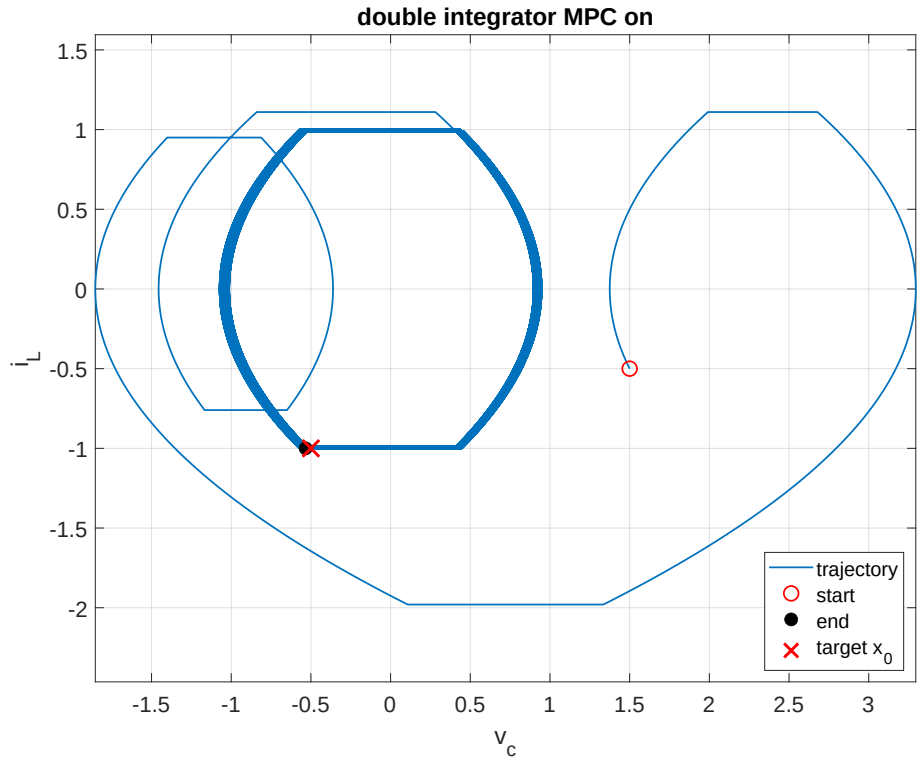


FIGURE 4.4 – State Trajectory MPC on, 50 cycles

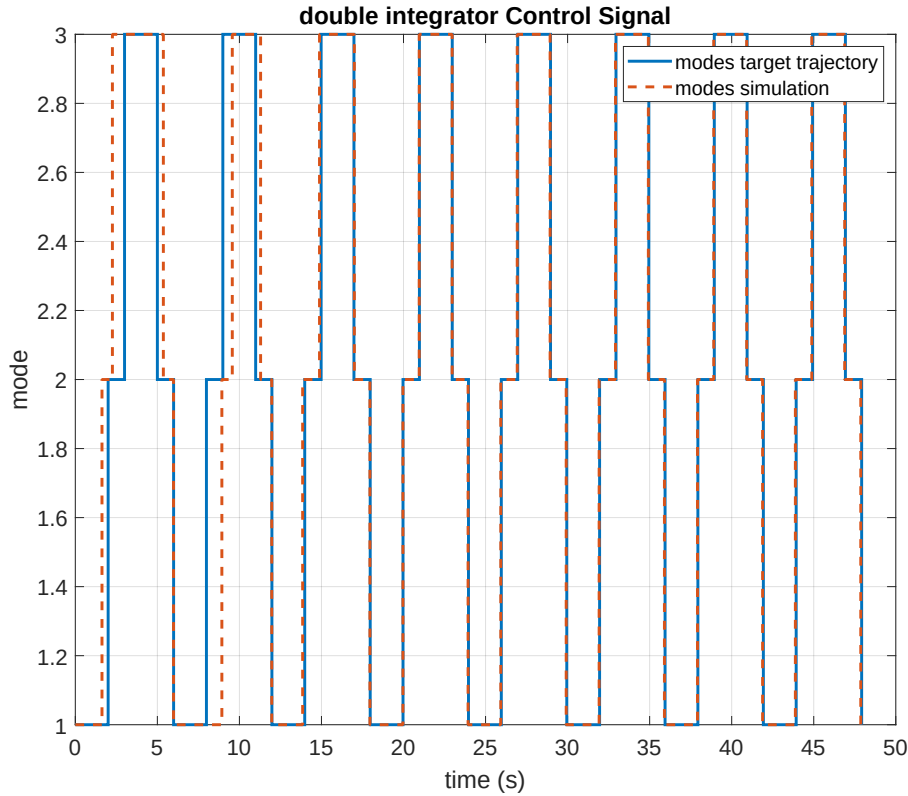


FIGURE 4.5 – Operation mode switch

4.4.2 Buck-boost converter

4.4.2.1 Parameters of the Simulation

This example describes the Buck-boost converter model described in session 2.3.1, the associated mode sequence for the target cyclic trajectory is shown in Figure 4.6

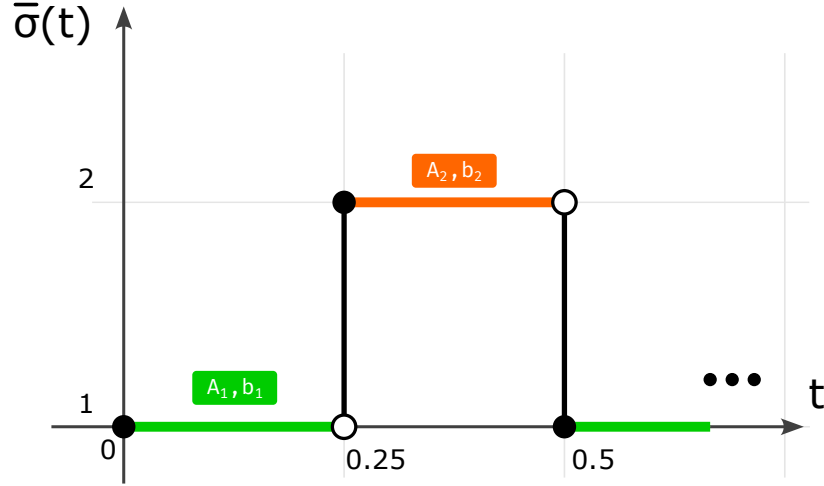


FIGURE 4.6 – Reference sequence for Buck-boost converter

where

$$A_1 = \begin{bmatrix} 0 & 0 \\ 0 & -1/(RC) \end{bmatrix}, \quad A_2 = \begin{bmatrix} 0 & 1/L \\ -1/C & -1/(RC) \end{bmatrix}$$

$$b_1 = \begin{bmatrix} E/L \\ 0 \end{bmatrix} \quad b_2 = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

and the operation mode is characterized by the index $\Omega = [1, 2]$. The time switch values are defined as $\tau^\infty = [0, 0.2513, 0.5914]$, and the reference state vector for this trajectory is denoted as $x_{ref} = [2, -1]$. Additionally, the maximum period value for the trajectory cycle is specified as $T_{pmax} = 1$, while the initial state of the nominal target trajectory is given by $x_0 = [1.8708, -1.1198]$. The time step of the simulation is $t_{step} = 1 \times 10^{-5}$ and the weight matrix used in the computation of the trajectory is Q is a diagonal matrix of ones.

4.4.2.2 Prediction Matrices

The prediction matrices are constructed using (4.32) and (4.33), and the result is

$$\Phi = \begin{bmatrix} 0.9713 & 0.2192 \\ -0.1707 & 0.5857 \end{bmatrix} \quad (4.58)$$

$$\Gamma = \begin{bmatrix} 2.4556 \\ 0.9280 \end{bmatrix} \quad (4.59)$$

with the switching time constraints

$$c = \begin{bmatrix} -\Delta t_{r_1} + \Delta t_{r_1, \min} \\ -\Delta t_{r_2} + \Delta t_{r_2, \min} \\ \vdots \\ -\Delta t_{r_{N-1}} + \Delta t_{r_{N-1}, \min} \\ -\Delta t_{r_N} + \Delta t_{r_N, \min} \end{bmatrix} = \begin{bmatrix} -0.25 + 0.25 \\ -0.2504 + 0.25 \end{bmatrix} = \begin{bmatrix} 0 \\ -0.04 \end{bmatrix} \quad (4.60)$$

4.4.2.3 Results

Figure 4.6 shows the simulation of the trajectory for the system, in blue we see the trajectory without using the MPC controller, and in orange we see the trajectory using the MPC controller. They are both starting from $x_0 = [2.0493, -0.6108]$ and converge to the initial position of the target cyclic trajectory $\bar{x}_0 = [1.8493, -1.1108]$. Note that the trajectory with the MPC controller converges faster to the cyclic trajectory.

Figure 4.8 shows the control signal sequence, where we can see that the difference between the nominal target trajectory and the controlled trajectory decrease as they get in sync with each other.

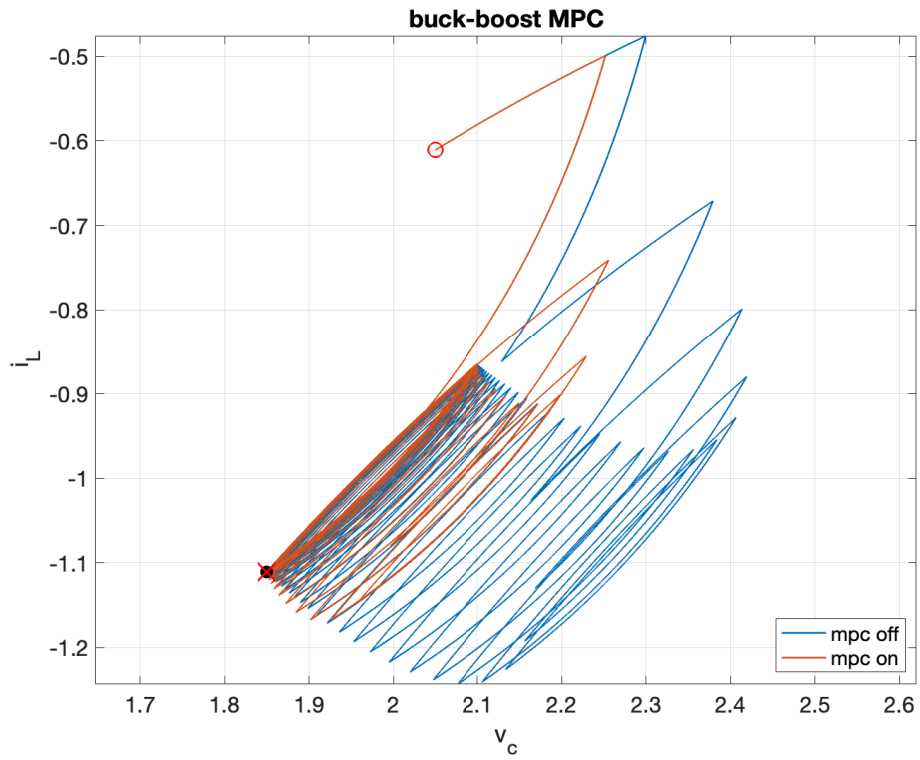


FIGURE 4.7 – State Trajectory for Buck-Boost converter

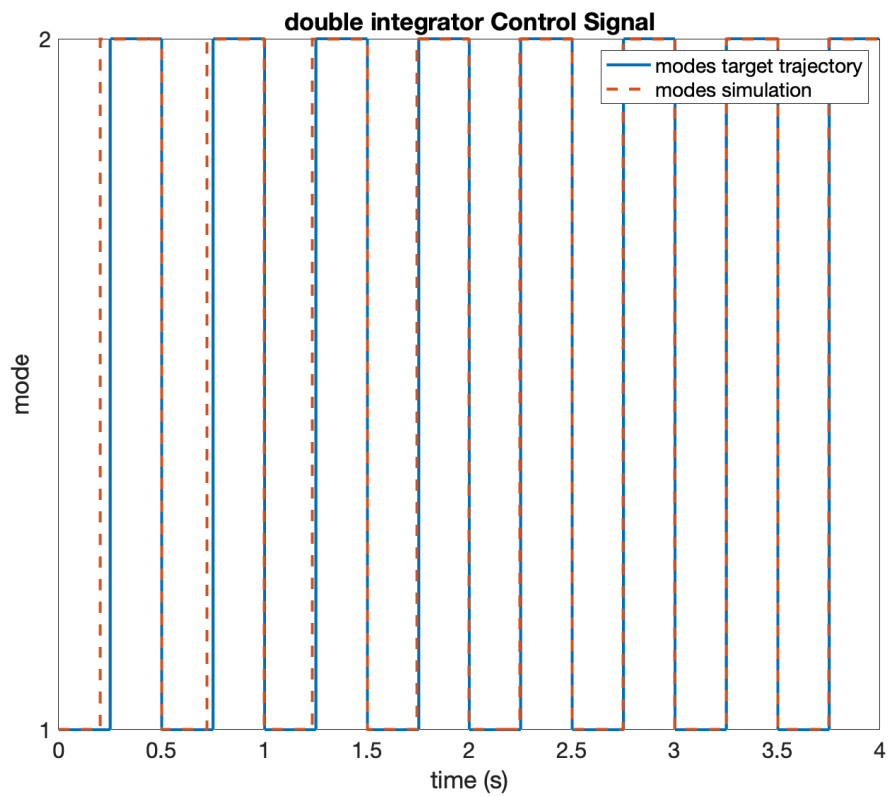


FIGURE 4.8 – Control Signal Sequence for Buck-Boost converter

4.4.3 Multilevel converter

4.4.3.1 Parameters of the Simulation

This example describes the Multilevel converter model described in session 2.3.2, the associated mode sequence for the target cyclic trajectory is shown in Figure 4.9.

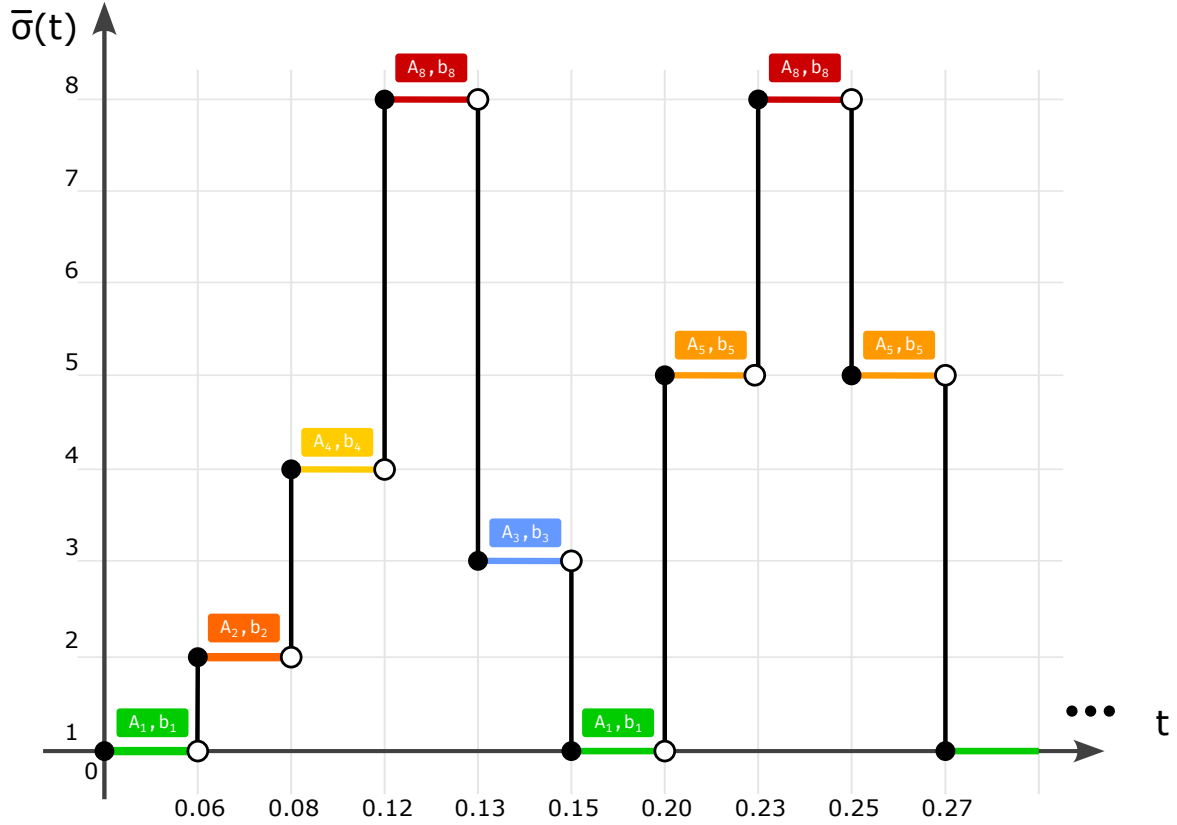


FIGURE 4.9 – Reference sequence for Multilevel converter

where

$$A_i = \begin{bmatrix} 0 & 0 & \begin{pmatrix} u_2(i) - u_1(i) \\ u_3(i) - u_2(i) \end{pmatrix} / C1 \\ 0 & 0 & \begin{pmatrix} u_2(i) - u_3(i) \\ -R/L \end{pmatrix} / L \end{bmatrix} \quad (4.61)$$

$$b_i = \begin{bmatrix} 0 \\ 0 \\ (E/L)u_3(i) \end{bmatrix}$$

with $R = 10.0\Omega$, $L = 10.0mH$, $C_1 = 40.0\mu F$, $C_2 = 40.0 \times 10^{-6}\mu F$, $E = 30.0V$

and $u(i)$ is

TABLE 4.1 – Values of u_i for the Multilevel converter

mode (i)	u_1	u_2	u_3
1	0	0	0
2	0	0	1
3	0	1	0
4	0	1	1
5	1	0	0
6	1	0	1
7	1	1	0
8	1	1	1

and the operation mode is characterized by the index $\Omega = [1, 2, 4, 8, 3, 1, 5, 8, 5]$. The time switch values are defined as $\tau^\infty = [0, 0.0639, 0.0878, 0.1119, 0.1344, 0.1567, 0.2065, 0.23, 0.2529, 0.2756]$, and the reference state vector for this trajectory is denoted as $x_{ref} = [10, 20, 1]$. Additionally, the maximum period value for the trajectory cycle is specified as $T_{p_{max}} = 4 \times 10^{-4}$, while the initial state of the nominal target trajectory is given by $x_0 = [9.5143, 19.8211, 1.0314]$. The time step of the simulation is $t_{step} = 1 \times 10^{-7}$ and the weight matrix used in the computation of the trajectory is $Q = diag([10, 5, 20000])$.

4.4.3.2 Prediction Matrices

The prediction matrices are constructed using (4.32) and (4.33), and the result is

$$\Phi = \begin{bmatrix} 0.999539 & 0.001900 & -0.124610 \\ -0.002031 & 0.999915 & 0.000764 \\ -0.000461 & -0.000290 & 0.759578 \end{bmatrix} \quad (4.62)$$

$$\Gamma = \begin{bmatrix} 0.007784 & -2.578391 & -0.081618 \\ -2.400410 & 2.472097 & -0.079094 \\ 2.477894 & 0.042370 & -0.076050 \\ -2.602774 & 2.388977 & 0.171509 \\ 2.472432 & -2.594318 & 0.102211 \\ 2.703328 & 0 & -0.087828 \\ -2.354659 & 0 & -0.189865 \\ 2.354483 & 0 & 0.199932 \end{bmatrix}^T * \times 10^4 \quad (4.63)$$

with the switching time constraints

$$c = \begin{bmatrix} -\Delta t_{r_1} + \Delta t_{r_1, \min} \\ -\Delta t_{r_2} + \Delta t_{r_2, \min} \\ \vdots \\ -\Delta t_{r_{N-1}} + \Delta t_{r_{N-1}, \min} \\ -\Delta t_{r_N} + \Delta t_{r_N, \min} \end{bmatrix} = \begin{bmatrix} -0.638811 + 0.20 \\ -0.239318 + 0.20 \\ -0.240440 + 0.20 \\ -0.225094 + 0.20 \\ -0.223339 + 0.20 \\ -0.497696 + 0.20 \\ -0.235714 + 0.20 \\ -0.228086 + 0.20 \\ -0.227575 + 0.20 \end{bmatrix} \times 10^{-4} = \begin{bmatrix} -0.438811 \\ -0.039318 \\ -0.040440 \\ -0.025094 \\ -0.023339 \\ -0.297696 \\ -0.035714 \\ -0.028086 \\ -0.027575 \end{bmatrix} \times 10^{-4} \quad (4.64)$$

4.4.3.3 Results

Figure 4.10 shows the trajectory for the Multilevel converter, in blue we see the trajectory without using the MPC controller, and in orange we see the trajectory using the MPC controller. They are starting from $x_0 = [7.5143, 20.8211, 0.0314]$, the trajectory with the MPC off goes to a point far from the targeted position $\bar{x}_0 = [9.5143, 19.8211, 1.0314]$, but the trajectory with the MPC on goes drives the system to the desired target.

Figure 4.14 displays the control signal sequence, we can see the difference between the open loop and the controlled signal decreasing and getting in sync.

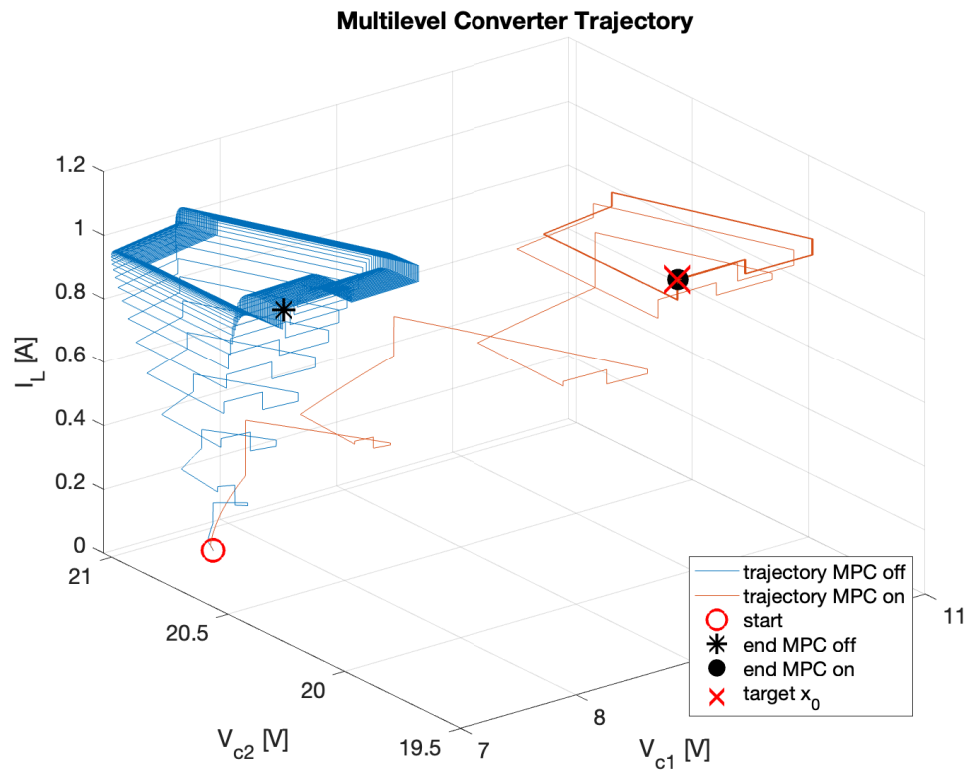


FIGURE 4.10 – State Trajectory for Multilevel converter

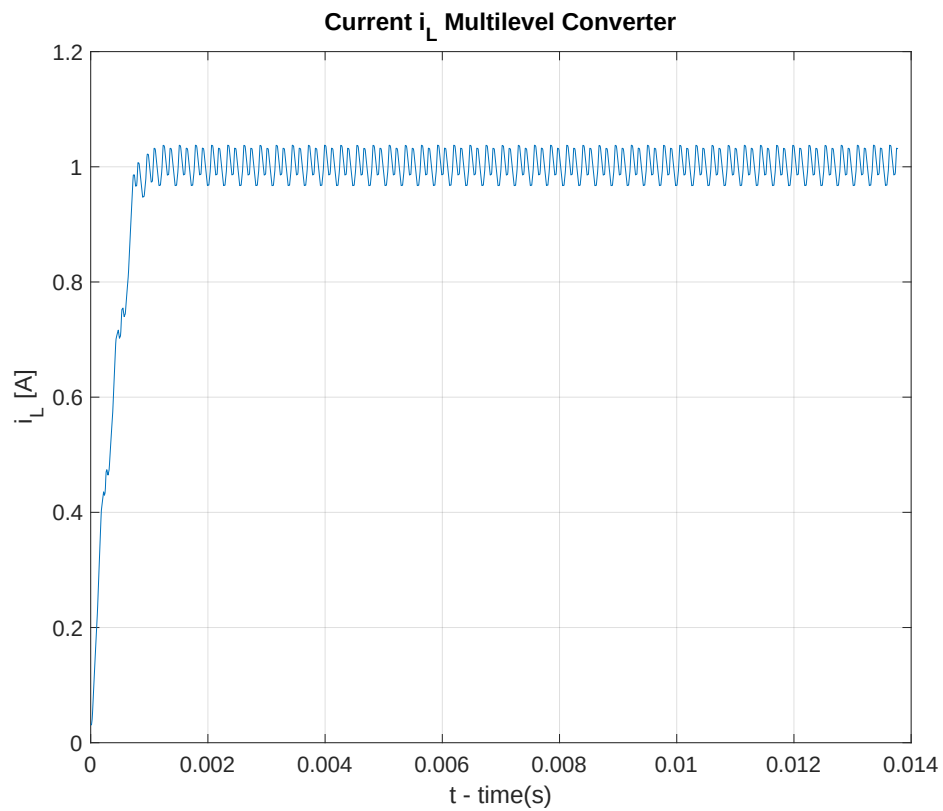


FIGURE 4.11 – fig 3

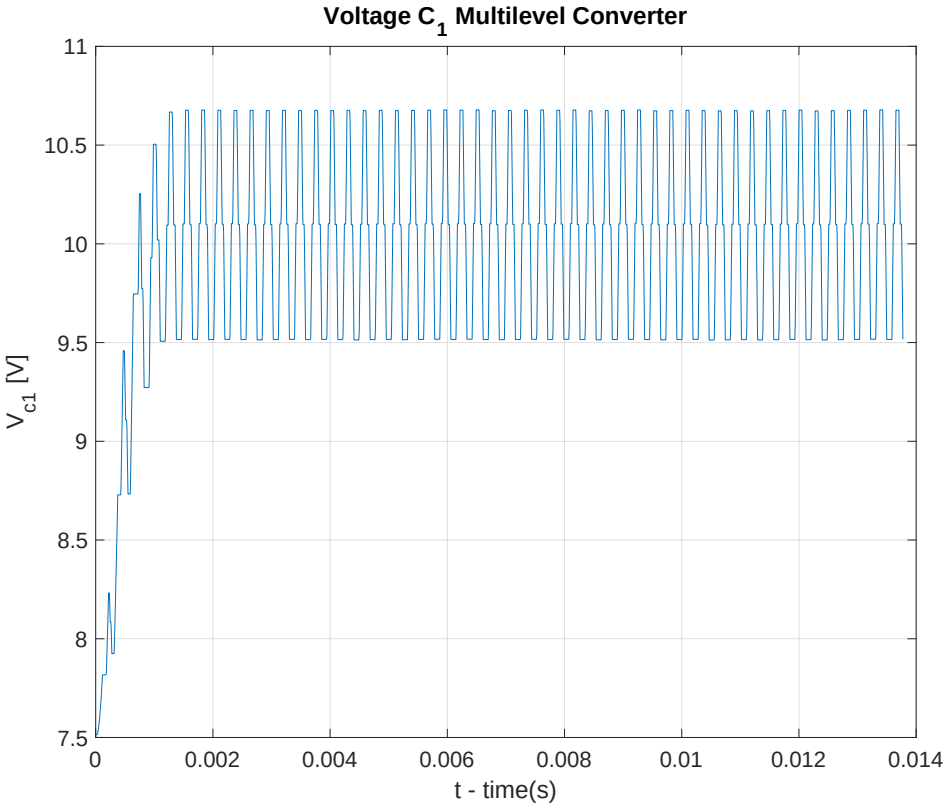


FIGURE 4.12 – fig 1

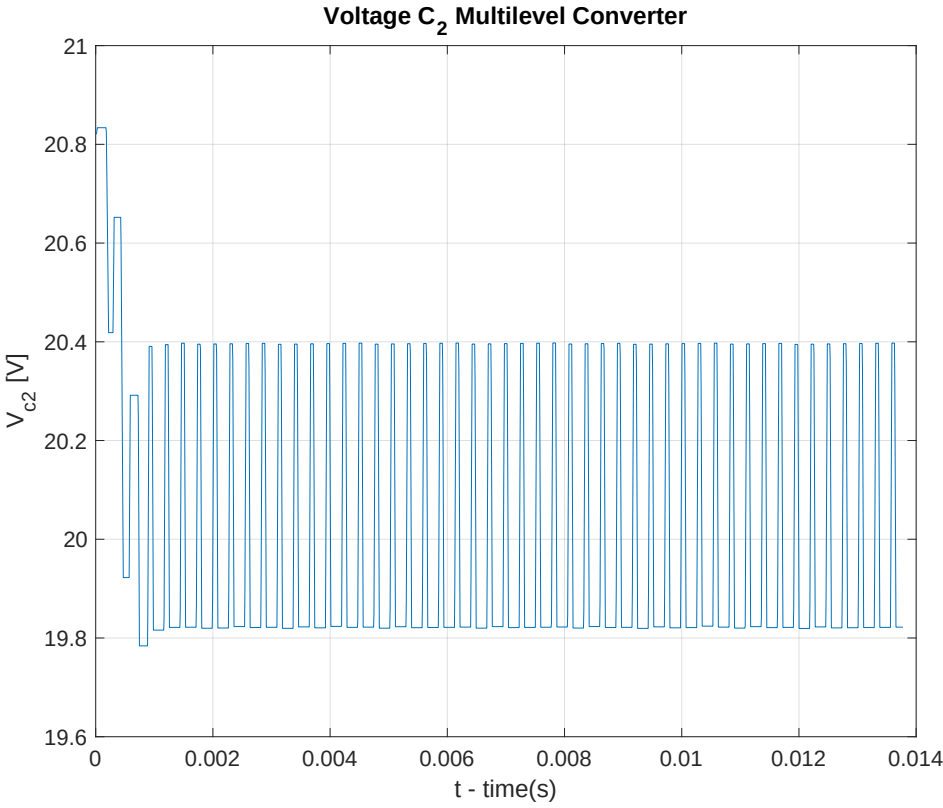


FIGURE 4.13 – fig 2

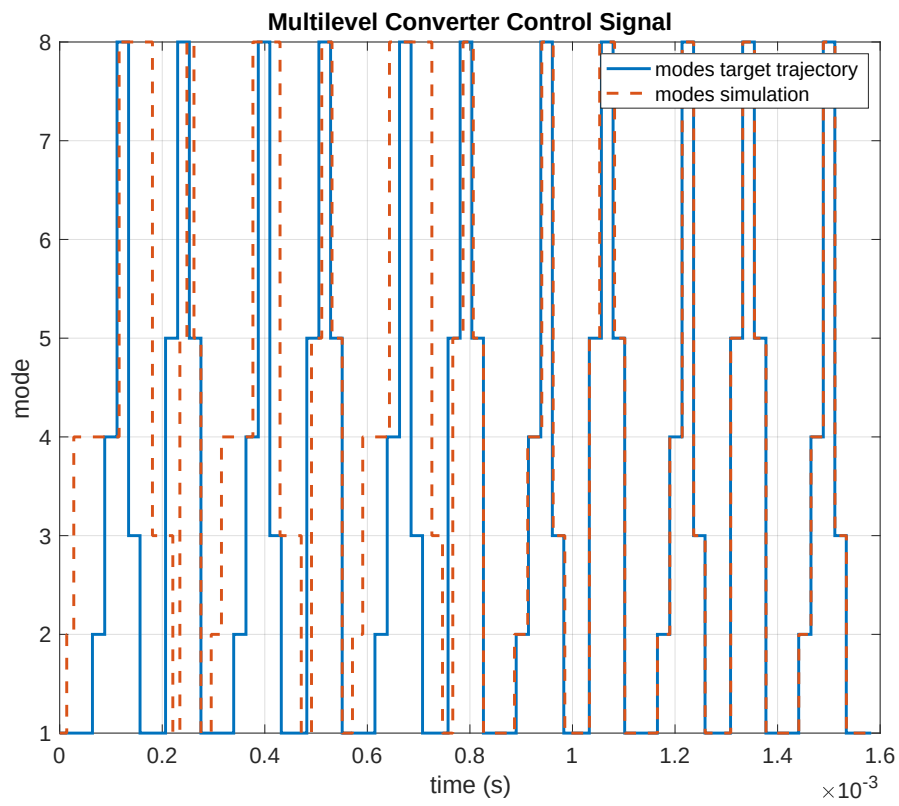


FIGURE 4.14 – Variant Dynamic Circuit 2: State Trajectory

4.5 Conclusion

5 Conclusão

- conclusoes dos resultados apresentados
- plano de trabalho para conclusao do doutorado
- cronograma e atividades para conclusao do doutorado
- trabalhos finais: deducacao alternativa do estado aumentado

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Appendix A - Tópicos de Dilema Linear

A.1 Modelo de simulação

```
+engine
├─ CORES.m
├─ fun_custo_patino.m
├─ get_config_sim_matheus.m
├─ get_config_sim_patino_1.m
├─ get_config_sim_patino_2.m
├─ get_config.m
├─ get_otmin_opt.m
├─ get_ts.m
├─ get_x0.m
├─ get_xr.m
├─ nonlincon_patino.m
├─ otmin.m
├─ sim_1.m
├─ sim_n.m
└─ +mpc
    ├─ construcao_modelo_instantes.m
    ├─ matrizes_ss_mpc_dualmode_switching.m
    └─ mpc_dualmode_switching.m
```

A.1.1 get_x0

funcao de calculo da condicao inicial

A.1.2 get_xf

Texto do modelo de simulação sendo colocado aqui.

Annex A - Exemplo de um Primeiro Anexo

A.1 Uma Seção do Primeiro Anexo

Algum texto na primeira seção do primeiro anexo.

FOLHA DE REGISTRO DO DOCUMENTO

1. CLASSIFICAÇÃO/TIPO TD	2. DATA 25 de março de 2015	3. DOCUMENTO Nº DCTA/ITA/DM-018/2015	4. Nº DE PÁGINAS 69
5. TÍTULO E SUBTÍTULO: Predictive control of switched affine systems subjected to dwell time constraints			
6. AUTOR(ES): Daniel Vieira			
7. INSTITUIÇÃO(ÕES)/ÓRGÃO(S) INTERNO(S)/DIVISÃO(ÕES): Instituto Tecnológico de Aeronáutica – ITA			
8. PALAVRAS-CHAVE SUGERIDAS PELO AUTOR: Cupim; Cimento; Estruturas			
9. PALAVRAS-CHAVE RESULTANTES DE INDEXAÇÃO: Cupim; Dilema; Construção			
10. APRESENTAÇÃO: (X) Nacional () Internacional ITA, São José dos Campos. Curso de Mestrado. Programa de Pós-Graduação em Engenharia Aeronáutica e Mecânica. Área de Sistemas Aeroespaciais e Mecatrônica. Orientador: Prof. Dr. Adalberto Santos Dupont. Coorientadora: Prof ^a . Dr ^a . Doralice Serra. Defesa em 05/03/2015. Publicada em 25/03/2015.			
11. RESUMO: Atuadores chaveados são difíceis de lidar por conta de algumas restrições características de sua natureza, porém eles são eficientes em inúmeras aplicações da engenharia, sendo aplicado, por exemplo, em sistemas de controle de atitude de veículos espaciais. Assim, saber utilizar desse tipo de atuador se faz importante e necessário. O principal problema a ser resolvido nessa tese é o projeto de um controlador que garanta a robustez de sistema com atuadores chaveados submetidos a restrições temporais que limitam a janela de resposta de seus comandos. Nessa abordagem, é feita uma busca de trajetória alvo de ciclo limite e, então, é calculado o erro dos estados e aplicada a técnica de controle preditivo para correção do erro e levar o systema à trajetória cíclica alvo.			
12. GRAU DE SIGILO: (X) OSTENSIVO () RESERVADO () SECRETO			